

4	C	
5	A	Change in momentum = area under F-t graph Area under F-t graph from 0 to 10 seconds = $0.5 \times (8+4) \times 12 + [(-0.5) \times (2) \times (12)] = 60 \text{ N s}$ Change in momentum = final momentum – initial momentum $60 = mv - 0$ $v = 200 \text{ m s}^{-1}$
6	A	Force by wall on water = $\rho \Delta V \times a$ = mass per unit time $\times \Delta v$